Data Science-240 Hours

S.No	Module Name	Topics	No. of Hours
1	**Module 1: Python & Data Foundations**	Python Basics: Syntax, Variables, Data Types, Control Flow, Functions	3
2		Python Collections: Lists, Tuples, Sets, Dictionaries & Comprehensions	2
3		Object-Oriented Programming (OOP): Classes, Objects, Inheritance, Encapsulation	3
4		NumPy: Arrays, Vectorized Operations, Broadcasting, Linear Algebra	3
5		Pandas: Series, DataFrames, Indexing, Merging, GroupBy, Aggregation	3
6		Data Cleaning: Handling Missing Values, Duplicates, Outliers, Type Conversion	3
7		File I/O: CSV, JSON, Excel; Working with APIs & Requests Library	3
8		Hands-On Lab: Retail Sales Analysis — Data Loading, Cleaning & Summarization	3
9		Hands-On Lab: Sales Performance Dashboard (Mini Project 1)	3
11	**Module 2: Data Collection & Engineering**	Web Scraping: BeautifulSoup, Scrapy, Selenium, Ethical & Legal Considerations	6
12		REST APIs & JSON Parsing: Requests, Authentication, Rate Limiting	6
13		Databases: SQL Fundamentals, SQLite, PostgreSQL, SQLAlchemy ORM	6
14		Data Pipeline Design: ETL Concepts, Workflow Orchestration Intro	6
15		Data Formats & Storage: Parquet, Feather, HDF5, Cloud Storage Basics	6
16		Hands-On Lab: Financial News + Stock Data Pipeline (Case Study)	3
17		Mini Project 2: Market Data Pipeline — Build End-to-End Data Ingestion Flow	3
19	**Module 3: Exploratory Data Analysis & Visualization**	EDA Foundations: Descriptive Stats, Distributions, Skewness, Kurtosis	6
20		Matplotlib: Line, Bar, Histogram, Scatter, Subplots, Custom Styling	6
21		Seaborn: Statistical Plots — Heatmaps, Pairplots, Violin, Box, FacetGrid	6
22		Interactive Visualization: Plotly & Dash — Dashboards & Drill-Down Charts	6
23		Correlation Analysis, Feature Relationships & Outlier Detection Methods	3

24		Hands-On Lab: Hospital Patient Readmission EDA (Case Study)	3
25		Mini Project 3: EDA & Visual Story — Social Media Dataset	3
26		Storytelling with Data: Narrative Techniques, Color Theory, Chart Selection	3
28	**Module 4: Machine Learning — Supervised Learning**	ML Fundamentals: Bias-Variance Tradeoff, Train/Val/Test Split, Overfitting	3
29		Linear & Logistic Regression: Math Intuition, Sklearn Implementation	3
30		Decision Trees & Random Forests: Gini, Entropy, Feature Importance	3
31		Gradient Boosting: XGBoost, LightGBM, CatBoost — Theory & Tuning	3
32		Support Vector Machines (SVM): Kernels, Hyperplanes, Margin Optimization	3
33		Model Evaluation: Confusion Matrix, ROC-AUC, Precision-Recall, F1-Score	3
34		Cross-Validation, Hyperparameter Tuning: GridSearchCV, RandomizedSearch	3
35		Feature Engineering: Encoding, Scaling, Polynomial Features, Selection	3
36		Hands-On Lab: Loan Default Risk — LendingClub Dataset (Case Study)	3
37		Mini Project 4: Credit Risk Classifier — End-to-End ML Pipeline	3
39	**Module 5: Machine Learning — Unsupervised Learning**	Clustering Algorithms: K-Means, Hierarchical, DBSCAN — Theory & Application	3
40		Dimensionality Reduction: PCA, t-SNE, UMAP — Intuition & Use Cases	3
41		Association Rule Mining: Apriori, FP-Growth — Market Basket Analysis	3
42		Anomaly Detection: Isolation Forest, One-Class SVM, Autoencoders Intro	3
43		Hands-On Lab: E-Commerce Customer Segmentation (Case Study)	3
44		Mini Project 5: Customer Segmentation Engine — RFM + Clustering	3
46		Cluster Evaluation: Silhouette Score, Davies-Bouldin, Elbow Method	3
47	**Module 6: Advanced ML & Model**	Ensemble Methods: Bagging, Boosting, Stacking, Voting Classifiers	3
48		Imbalanced Data: SMOTE, Class Weights, Threshold Tuning, ADASYN	3
49		Feature Selection: Recursive Feature Elimination (RFE), SHAP, LIME	3
50		Pipeline Architecture: Sklearn Pipelines, ColumnTransformer, Custom Transformers	3

51	**Module 6: Advanced ML & Model Engineering**	Model Explainability: SHAP Values, Feature Importance Visualization	3
52		Hands-On Lab: Churn Model Improvement — 72% → 88% AUC (Case Study)	3
53		Advanced Hyperparameter Optimization: Optuna, Bayesian Optimization	3
54		Major Project 1 Prep: Hospital Readmission Prediction (Problem Framing)	3
56	**Module 7: Deep Learning Foundations**	Neural Networks: Perceptrons, Activation Functions, Backpropagation, Gradient Descent	4
57		Convolutional Neural Networks (CNN): Architecture, Filters, Pooling, Applications	4
58		Recurrent Neural Networks (RNN) & LSTM: Time-Series, Sequential Data	6
59		Transfer Learning: Pretrained Models, Fine-Tuning — MobileNetV2, ResNet	3
60		TensorFlow & Keras: Model Building, Callbacks, Early Stopping, Saving	3
61		Hands-On Lab: Chest X-Ray Classification with Transfer Learning (Case Study)	3
62		Hands-On Lab: Image Classifier with MobileNetV2 — End-to-End Pipeline	3
64	**Module 8: Model Deployment & MLOps Basics**	Model Serialization: Pickle, Joblib, ONNX; Versioning Best Practices	3
65		Flask API Development: Building REST Endpoints, Request Handling, JSON Responses	3
66		Docker: Containerization, Dockerfile, docker-compose for ML Apps	3
67		CI/CD for ML: GitHub Actions, Automated Testing, Model Validation Pipelines	3
68		Cloud Deployment: AWS SageMaker / GCP Vertex AI Basics — Model Hosting	3
69		Hands-On Lab: Loan Approval API with Flask + Docker (Case Study)	3
70		Major Project 2 Prep: E-Commerce Churn Platform (Architecture Design)	2
72		Sprint 1: Problem Understanding, EDA, Baseline Modelling — Hospital Readmission	3

73	**Module 9: Capstone Industry Projects**	Sprint 2: Feature Engineering & Advanced Modelling — Hospital Readmission	3
74		Sprint 3: Model Evaluation, SHAP Analysis & Report — Hospital Readmission	3
75		Sprint 4: Deployment & API Integration — Hospital Readmission (Major Project 1)	3
76		Sprint 5: Problem Framing, Data Pipeline — E-Commerce Churn Platform	3
77		Sprint 6: Model Building & Explainability — E-Commerce Churn Platform	3
78		Sprint 7: Live Deployment, Dashboard & Revenue Impact — Churn Platform (MP2)	3
		Total No .of Hours	**240**